GRADUATION PROJECT

Road Damage Detection

Project Planning & Management

Model: YOLO11 object detection on the RDD2022 dataset

Date: October 2026

Student ID	Students Name
21031332	Mohamed Ahmed Ibrahim Hassan
21032696	Aliaa Mohamed Abdo Barakat
21124056	Youssef Mohamed Abdelhamid Zidane
21157939	Hany Ashraf Elsayed Mohamed

1. Project Proposal

The project builds an object-detection model that finds and classifies five types of road damage in photographs, so road-maintenance teams can survey pavement faster than by manual inspection. The baseline YOLO11n model already reaches 57.5% mAP@50 on the validation set.

0.575	0.620	0.543	4.3 ms
mAP@50	Precision	Recall	Inference / image

> Baseline results of YOLO11n (50 epochs, 640 px) on the validation split of 5,758 images, measured on a Tesla T4 GPU.

1.1 Overview

Roads deteriorate continuously, and manual surveys are slow, subjective and costly. We train a YOLO11 detector on the RDD2022 dataset (38,385 images split into train, validation and test) to locate each damage instance with a bounding box and a class label.

1.2 Objectives

- Train a detector for five classes: Longitudinal Crack, Transverse Crack, Alligator Crack, Other Corruption and Pothole.
- Reach at least 0.65 mAP@50 and 0.50 recall on Pothole, up from 0.575 and 0.37 in the baseline.
- Keep inference near real time (the baseline runs at about 4.3 ms per image on a Tesla T4).
- Package the best model so a user can upload an image and receive the annotated result.

1.3 Scope

In scope	Out of scope
2D detection on still images from RDD2022	Damage severity or repair-cost estimation
Five damage classes with bounding boxes	GPS mapping of damage on a road network
Training, evaluation, error analysis, model comparison	Live video from vehicle-mounted hardware
Simple demo application	Production hosting and large-scale deployment

2. Project Plan

The remaining work runs 8 weeks, from 15 October to 13 December 2026. Dataset preparation, exploration, baseline training and validation are already done.

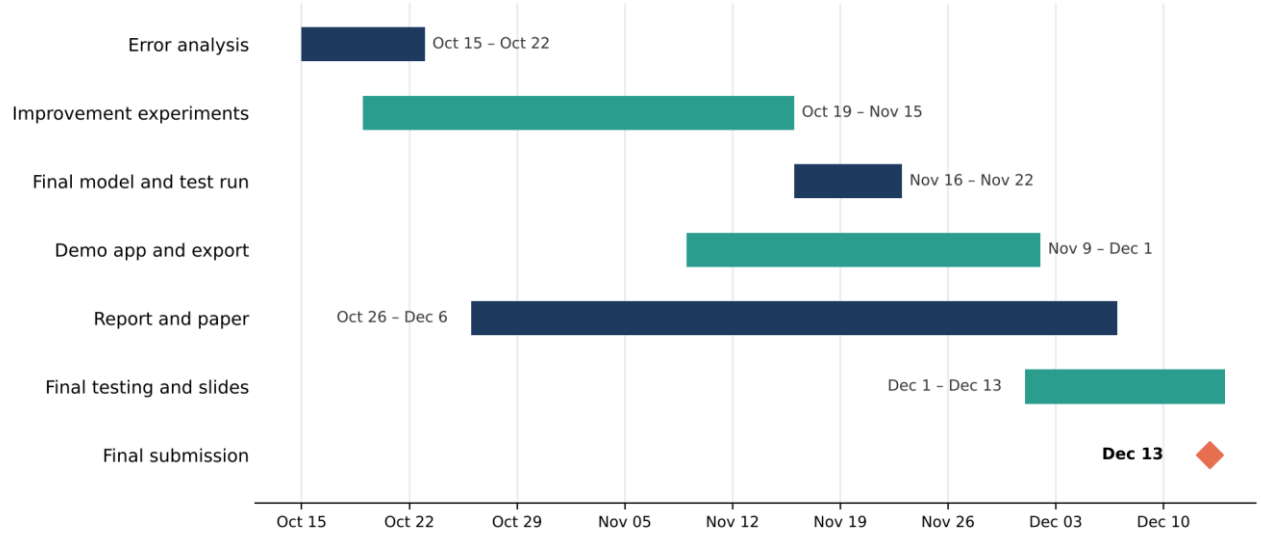


Figure 1 – Gantt chart of the remaining work packages

Experiments, the demo app and the report run in parallel after error analysis, and the final model is frozen on 22 November.

2.1 Milestones and deliverables

Milestone	Target	Deliverable
M0 Baseline model	Done	best.pt and validation metrics (0.575 mAP@50)
M1 Error analysis	22 Oct	Confusion-matrix review, failure-case gallery, baseline on the test split
M2 Experiments complete	15 Nov	Experiment log comparing model size, image size and Pothole fixes
M3 Final model frozen	22 Nov	Final weights and test-split metrics
M4 Demo ready	1 Dec	Exported model and demo app
M5 Final submission	13 Dec	Report, ML-course paper, slides

2.2 Resources

- **Compute:** Kaggle Tesla T4 GPU, the same setup as the baseline run.
- **Software:** Ultralytics YOLO11, PyTorch, OpenCV, pandas, seaborn.
- **Data:** RDD2022 Kaggle split (26,869 / 5,758 / 5,758 images).

3. Task Assignment & Roles

Four roles cover the project. Data preparation and the baseline are finished, so the open work is improvement, deployment and documentation.

Owner	Responsibilities	Main deliverables
[Mohamed]	Plan, meetings, risk log, KPI tracking.	Project plan, weekly status
[Aliaa]	Dataset split, label checks (duplicate labels, empty images), YAML config, augmentation	Clean train/Val/test split, data.yaml
[Youssef]	Training YOLO11, hyperparameter and model-size experiments	best.pt weights, experiment log
[Hany]	Report, paper, slides, final demo script	Final report and presentation

> Replace each [Name] with the team member who owns the role.

4. Risk Assessment & Mitigation Plan

The biggest technical risk is weak Pothole detection (recall 0.37 in the baseline). The biggest schedule risk is the December DEPI deadline against 6-hour training runs.

#	Risk	Likelihood	Impact	Mitigation
1	Low recall on Pothole (0.37) and weak Pothole mAP@50 (0.43)	**High**	**High**	Oversample Pothole images, tune class loss weight, try a larger model, lower the confidence threshold at inference
2	Class imbalance: Longitudinal Crack is 39% of training objects, Pothole 10%	**High**	**Medium**	Class-aware sampling, targeted augmentation, report per-class metrics, not only the overall average
3	Long training time: 50 epochs took 6.15 hours on a Tesla T4	**High**	**High**	Run experiments in parallel on separate Kaggle sessions, test on a data subset first, save checkpoints every run
4	Free GPU quota or session limits interrupt training	**Medium**	**High**	Resume from last.pt, keep a second account or Colab as backup, download weights after each run
5	Domain shift: images come from several countries, cameras and weather conditions	**Medium**	**Medium**	Evaluate per country where possible, add brightness and blur augmentation, keep a held-out test split

#	Risk	Likelihood	Impact	Mitigation
6	Label noise, such as the duplicate label found in Japan_006536.jpg	**Medium**	**Low**	Run a label audit script, remove duplicates and invalid boxes before retraining
7	Overfitting to the validation set through repeated tuning	**Medium**	**Medium**	Report final numbers on the untouched test split only
8	Team member unavailable during exams or final-year workload	**Medium**	**Medium**	Cross-train two people per task, set weekly milestones, keep the buffer in December
9	Demo fails on presentation day	**Low**	**High**	Record a backup video, test on a second machine, ship a lightweight exported model

5. KPIs (Key Performance Indicators)

The baseline meets the speed goal but not the accuracy goals: overall mAP@50 is 0.575 against a 0.65 target.

KPI	Baseline	Target	How it is measured
mAP@50 (all classes)	0.575	≥ 0.65	Ultralytics validation on the test split
mAP@50–95	0.311	≥ 0.38	Same run
Precision	0.620	≥ 0.68	Same run
Recall	0.543	≥ 0.60	Same run
Pothole recall	0.368	≥ 0.50	Per-class validation table
Inference time per image	4.3 ms (T4 GPU)	≤ 10 ms on GPU	Ultralytics speed report
Model size	5.5 MB, 2.58 M parameters	≤ 50 MB	Weight file size
Training time per experiment	6.15 h (50 epochs)	≤ 8 h	Training log
Demo response time	Not built yet	≤ 2 s per image	Timed in the demo app
Demo uptime during evaluation week	Not built yet	≥ 99%	Manual check log
Milestones delivered on time	Not tracked	≥ 90%	Project plan review

> Baselines come from the validation run in the notebook. The targets are proposed and should be agreed with the supervisor.